

Gourab Roy

AI Researcher & Erasmus Mundus IPCVAI Student

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Location: Bordeaux, France

Google Scholar: 25+ citations

RESEARCH INTERESTS

Computer vision, deep learning, Explainable AI (XAI), vision transformers, biomedical imaging (CT/MRI), few-shot learning, and trustworthy AI for real-world deployment.

EDUCATION

Erasmus Mundus Joint Master's Degree – IPCVAI

Artificial Intelligence for Image Processing and Computer Vision

Europe

2025–2027 (ongoing)

◦ Semester 3 – Université de Bordeaux (UBx), France

Starting Sep 2026

Coursework: Explainable AI for Images (XAI), AI & Sustainability, Advanced Computer Vision, Acquisition, Reconstruction & Medical Imaging.

◦ Semester 2 – Universidad Autónoma de Madrid (UAM), Spain

Completed Jun 2026

Coursework: HPC for Deep Learning, Deep Learning for Visual Signal Processing I, Deep Learning for Computer Vision II, Vision for Multiple/Moving Cameras, Tutored Research & Development (TRDP 1).

◦ Semester 1 – Pázmány Péter Catholic University (PPKE), Budapest, Hungary

Completed Jan 2026

Coursework: Basic Image Processing, Data Mining & Machine Learning, Fundamentals of Deep Learning, Multimodal Sensor Fusion & Navigation, Biomedical Signal Processing, Design Patterns.

Sylhet Engineering College

Sylhet, Bangladesh

B.Sc. Computer Science and Engineering; CGPA: 3.56 / 4.00

2018–2023

- **Undergraduate Thesis:** *CerebralNet*: A deep learning model for detecting major abnormalities in brain CT and MRI scans using segmentation and classification techniques.

SKILLS SUMMARY

- **Languages:** Python, C, C++, Java, JavaScript, MATLAB
- **Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNNs, Vision Transformers (ViT), YOLO (v11/v12), CLIP, Attention Rollout, transfer learning, few-shot learning
- **Computer Vision:** Image classification, segmentation, object detection, multi-view geometry, Explainable AI (XAI)
- **Data Analysis:** NumPy, Pandas, Matplotlib, SciPy, Jupyter Notebook
- **Tools & Platforms:** Git, MySQL, SQLite, Linux, Windows
- **Research:** L^AT_EX, academic writing, peer review (*Informatics in Medicine Unlocked*, Elsevier)

EXPERIENCE

Sylhet Engineering College, Department of CSE

Sylhet, Bangladesh

Research Assistant

Nov 2023 – Oct 2025

- **Applied ML Research:** Led model design, experimental evaluation, and paper preparation across medical AI, document AI, NLP, agricultural AI, and structural computer vision, contributing to 12 scholarly works.
- **Mentorship:** Mentored undergraduate students on final-year theses and research projects.
- **Supervisor:** Prof. Md. Abu Naser Mojumder, Head of CSE.

Sylhet Engineering College, Department of CSE

Sylhet, Bangladesh

Adjunct Lecturer

Jan 2024 – Aug 2025

- **Teaching:** Taught Distributed Computing, Computer Architecture, Introduction to Data Science, Peripheral Interfacing, and Microprocessors & Microcontrollers.
- **Academic Support:** Designed assessments, held consultation sessions, and co-supervised final-year thesis projects.

PUBLICATIONS – [Google Scholar](#)

Journal Articles

- [J1] “CovidExpert: A Triplet Siamese Neural Network framework for the detection of COVID-19,” *Informatics in Medicine Unlocked*, Elsevier, 2023. [doi:10.1016/j.imu.2022.101156](https://doi.org/10.1016/j.imu.2022.101156)
- [J2] “Image-based classification of Bangladeshi rice varieties using deep convolutional neural networks,” *Agricultural Products Processing and Storage*, Springer Nature, 2026. [doi:10.1007/s44462-026-00061-9](https://doi.org/10.1007/s44462-026-00061-9)
- [J3] “Advancing Bangla typography: machine learning and transfer learning based font detection and classification approach using the BanglaFont45 dataset,” *Journal of Computer Sciences Institute*, 2025. [doi:10.35784/jcsi.7120](https://doi.org/10.35784/jcsi.7120)
- [J4] “Optical character recognition for ancient scripts: a case study on Syloti Nagri using deep learning models,” *Journal of Computer Sciences Institute*, 2025. [doi:10.35784/jcsi.6785](https://doi.org/10.35784/jcsi.6785)

Conference Papers

- [C1] “PlantXNet: A CNN-Based Explainable AI Framework for Multi-Plant Leaf Disease Classification Using LIME,” *IEEE QPAIN 2026*. doi:10.1109/QPAIN69676.2026.11545914
- [C2] “MaizeXNet: A CNN-Based Explainable AI Framework for Maize Pest and Disease Classification,” *IEEE QPAIN 2026*. [link](#)
- [C3] “PoultryXNet: An Explainable Deep Learning Framework for Poultry Disease Classification From Fecal Images,” *IEEE QPAIN 2026*. doi:10.1109/QPAIN69676.2026.11545990
- [C4] “DateFNet: From Pixels to Plates – Attention Based Multi-Stream CNN for Date Fruit Classification,” *ICCIT 2025*, IEEE. doi:10.1109/ICCIT68739.2025.11491249
- [C5] “Deep Learning-Based Detection of Aircraft Cracks and Dents Using YOLOv11 and YOLOv12 for Aviation Safety in Bangladesh,” *ICCCNT 2025*, IEEE. [link](#)
- [C6] “Enhancing Concrete Performance: Machine Learning Models for Predicting and Classifying Tensile Strength,” *ECCE 2025*, IEEE. doi:10.1109/ECCE64574.2025.11013991
- [C7] “CrackClassification4: Classifying Four Types of Cracks in Structural Surfaces Using CNNs,” *ICCIT 2024*, IEEE. doi:10.1109/ICCIT64611.2024.11022574
- [C8] “Optimizing Multilingual Summarization: Advanced Corpus Creation and Filtering Approach,” *ICCIT 2024*, IEEE. doi:10.1109/ICCIT64611.2024.11022549

SELECTED PROJECTS

Cross-Modal Attention Bridge (CMAB) for CLIP Interpretability

UAM Madrid

TRDP 1 research project

Feb 2026 – Ongoing

- **Focus:** Explainable AI for CLIP image–text alignment; supervised by [Prof. Jenny Benois-Pineau](#) (UBx Bordeaux) and [Prof. Marcos Escudero-Viñolo](#) (UAM Madrid).
- **Contribution:** Developing K-sigma filtered attention rollouts to produce fine-grained, text-conditional visual explanations per token without modifying model weights.
- **Tools:** PyTorch, ViT, CLIP, Attention Rollout, XAI.

Cardiac Autonomic Assessment via PPG-Derived HRV

PPKE Budapest

Sound stimulation and physiological signal analysis

Dec 2025

- **Focus:** PPG-derived heart-rate variability analysis across baseline and music-stimulation conditions for 10 participants.
- **Contribution:** Built a modular Python pipeline with Butterworth filtering, adaptive peak detection, FFT spectral analysis, Friedman/Wilcoxon testing, and an individual responsiveness classifier.
- **Result:** All subjects showed significant autonomic nervous system shifts despite non-significant group means; [repository](#).
- **Tools:** Python, NumPy, SciPy, Pandas, Matplotlib.

Calibrated Multi-View 3D Reconstruction

UAM Madrid

Sparse reconstruction and bundle adjustment

2025

- **Focus:** Calibrated sparse 3D reconstruction from multiple camera views.
- **Contribution:** Implemented a MATLAB pipeline for camera calibration, SIFT/SURF/ORB feature matching with RANSAC, fundamental matrix estimation, epipolar geometry, reconstruction, and bundle adjustment.
- **Result:** Achieved 4.47 px reprojection error over 672 multi-view tracks; [repository](#).
- **Tools:** MATLAB, SIFT, RANSAC, Bundle Adjustment, Epipolar Geometry.

Data Analysis Projects

Exploratory data analysis and visualization

- **Focus:** Exploratory data analysis and visualization across Boston housing, COVID-19 global health outcomes, and Google Play app ratings datasets.
- **Contribution:** Created a GitHub portfolio of 8 EDA projects using data cleaning, visualization, and notebook-based reporting workflows.
- **Repository:** github.com/GourabRoy551/DataVisualization.
- **Tools:** Python, NumPy, Matplotlib, Pandas, Jupyter Notebook.

CERTIFICATIONS & ACHIEVEMENTS

- **Problem Solving:** Solved 350+ problems on [HackerRank](#), [Codeforces](#), and [UVA](#).
- **Achievements:** Earned gold badges in C++ and Python problem-solving on [HackerRank](#).
- **Machine Learning Specialization:** Completed the Machine Learning Specialization on Coursera [\[Link\]](#).
- **Data Structures Course:** Completed the Data Structures course on Coursera [\[Link\]](#).
- **Data Visualization Masterclass:** Completed the Data Visualization in Python Masterclass on Udemy [\[Link\]](#).
- **Review Contribution:** Contributed as a reviewer for *Informatics in Medicine Unlocked* [\[Link\]](#).